Space Mission Analysis

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Chair of Space Technology at a Glance

Small-satellite missions from first concept to in-orbit operations

21
missions to date

32
satellites launched and operated

~50
researchers

~25
student assistants

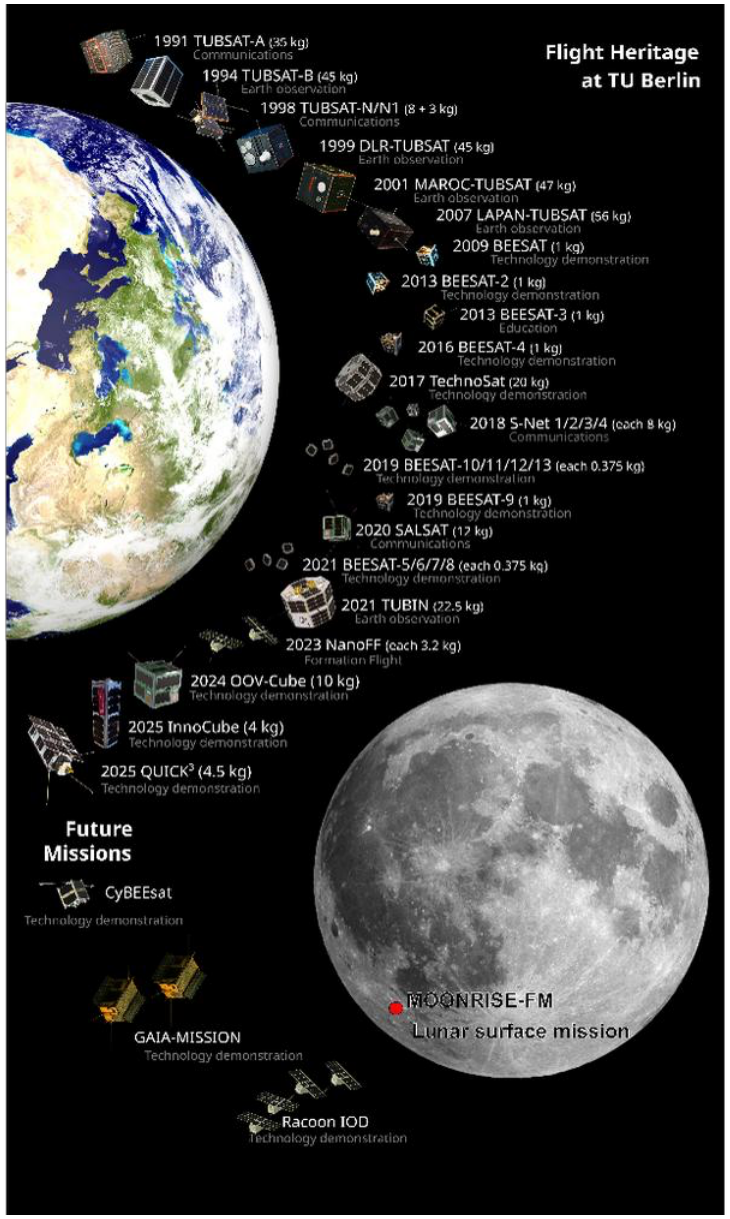
Three working groups:

Exploration & Propulsion
Regolith 3D printing, MOONRISE laser melting, propulsion test stands

Distributed Space Systems
Formation flight (NanoFF), constellations (S-NET), inter-satellite links

On-Orbit Servicing & Autonomy
Rendezvous & docking, robotic arms (NanoRnD), gecko grippers (GEICKO)

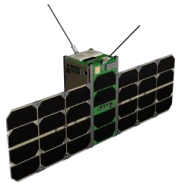
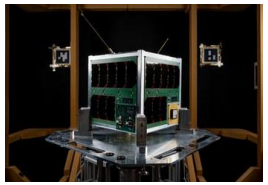
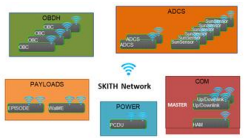
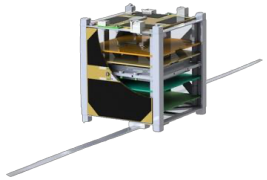
Infrastructure
Mission control centre · simulation & integration labs · propulsion test stands (1–50 N) · optical ground station TUBOGS (80 cm mirror, DLR Neustrelitz)



Satellite Missions 2023–2029

Four launches since 2023 – and four missions in the pipeline until 2029

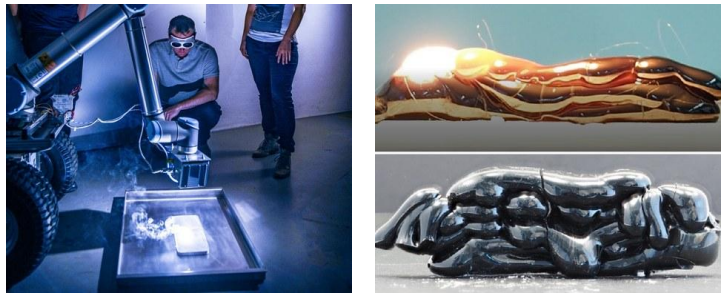
In orbit | **In development**

Dec 2023	Jul 2024	Jan 2025	Jun 2025	2026	2028	2028/29	End 2029
							
NanoFF Falcon 9 2 × 2U formation flight at < 2 km	**OOV-Cube** Ariane 6 maiden flight TUBiX10 (10 kg): L-band link to GEO, mioty IoT	**InnoCubE** SpaceX Transporter-12 Wireless data bus (SKITH), energy-storing structure	**QUICK³** SpaceX Transporter-14 Quantum photonics: single-photon source	**CyBEEsat** Isar Aerospace Spectrum 1U, 2 kg: cyber-secure RACCOON OS	**GAIA** 2 × 16U Wildlife monitoring via mioty IoT, Ka-band ISL	**RACOON** 3 × TUBiX5 (3U) RF geolocation, post-quantum secure comms	**OMLO IOD** project start 06/2026 Optical machine learning in orbit

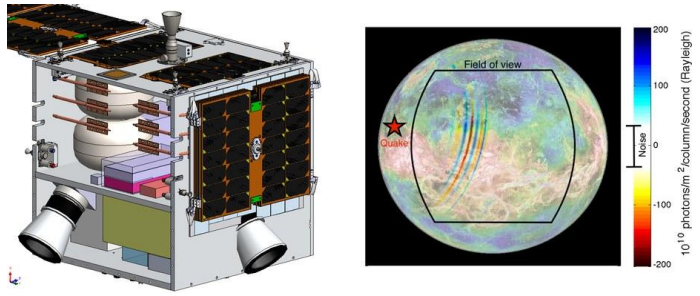
➔ TUBiX platform family scales from 3U (TUBiX5) and 10 kg (TUBiX10) to 60+ kg microsatellites (TUBiX20)

Exploration and On-Orbit Servicing

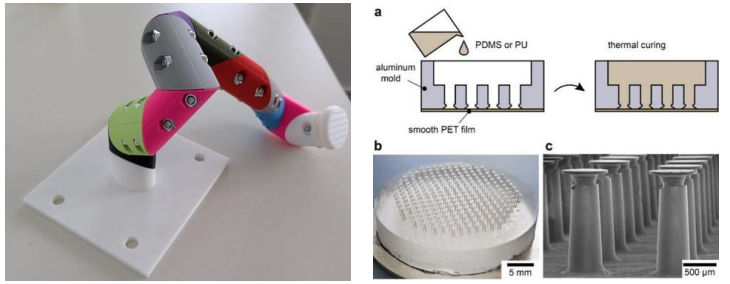
From lunar surface technology to interplanetary small satellites and servicing in orbit



MOONRISE melting regolith simulant · regolith melt



SER3NE concept · simulated Venus-quake airglow



Robotic mini-arm prototype · gecko-type adhesive pads

The Moon: building with regolith

- MOONRISE: laser melting on the lunar surface – flight booked with Astrobotic
- EDAM-R & 3D-LAVA: 3D printing with lunar regolith in vacuum

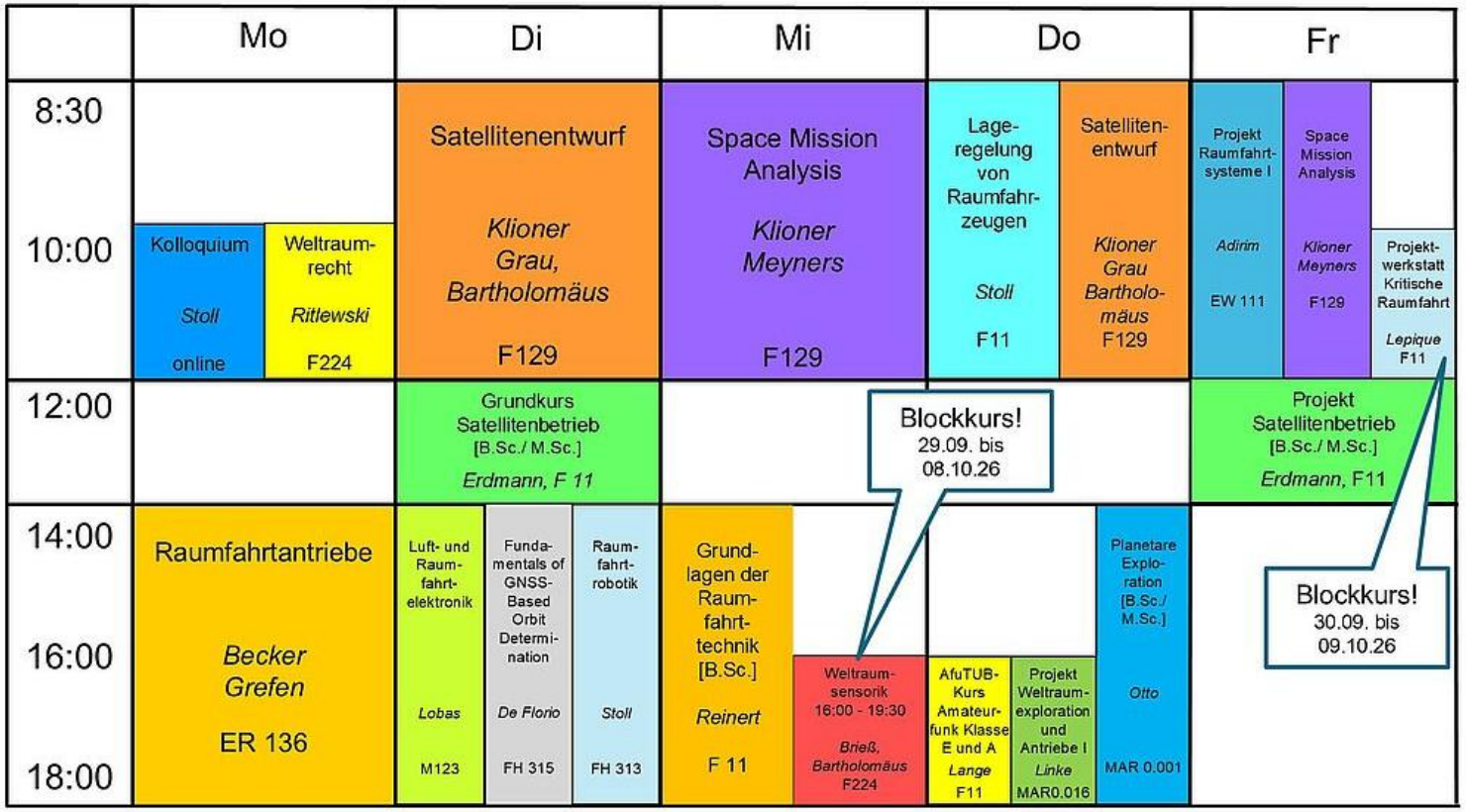
Interplanetary small satellites (TUBiX20)

- SER3NE: lunar orbiter mapping volatiles, geology and ISRU resources
- Quo Vadis: detecting Venus quakes from orbit via airglow

On-orbit servicing

- NanoRnD: 6U rendezvous & docking concept with 1U robotic arm (TRL 7)
- Gecko docking pads tested on the ISS; GEICKO capture kit (EIC Pathfinder)

Modules WS 26/27



Useful Links

- RFT InfoPoint

  https://isis.tu-berlin.de/course/view.php?id=25396

- Colloquium

  https://www.raumfahrttechnik.tu-berlin.de/menue/studium_und_lehre/lehrveranstaltungen/kolloquium_raumfahrttechnik/

What do you expect?

Content

- Intro and Requirements
- Orbit Propagation
- Numerical Methods
- Orbit Determination
- 3-Body Problem
- Interplanetary Transfer
- Trajectory Optimization
- Space Debris Mitigation Strategies
- Orbit Choice and Design

Organization

- Weekly slot Wednesday and Friday 9.00 – 12 am, F129: lectures and exercises
- Homework will be announced and discussed during exercises
- Grading: 50 % Assignments, 20% Paper, 30% Exam
- Assignments shall be submitted in groups of 2 to max. 3 students, groups can change from assignment to assignment
- **Exam: planned for 12.02.2027 9-10am, closed book**
  - Multiple choice
  - Open questions

Homework

- Coding
  - Respect coding guidelines
  - Write comprehensive code with comments where needed
  - Be careful to submit all relevant files
- Report
  - Keep it simple, no long stories but DO ADD DESCRIPTIONS
  - Write comprehensively
- Final Homework: Paper

An Introduction

Motivation for Mission Analysis

1. It‘s rocket science. At least part of it.
2. Planning is half the fun and more than half of a space mission.
3. Space missions take time and many loops to run.
4. Space is expensive.
5. It‘s somehow political.

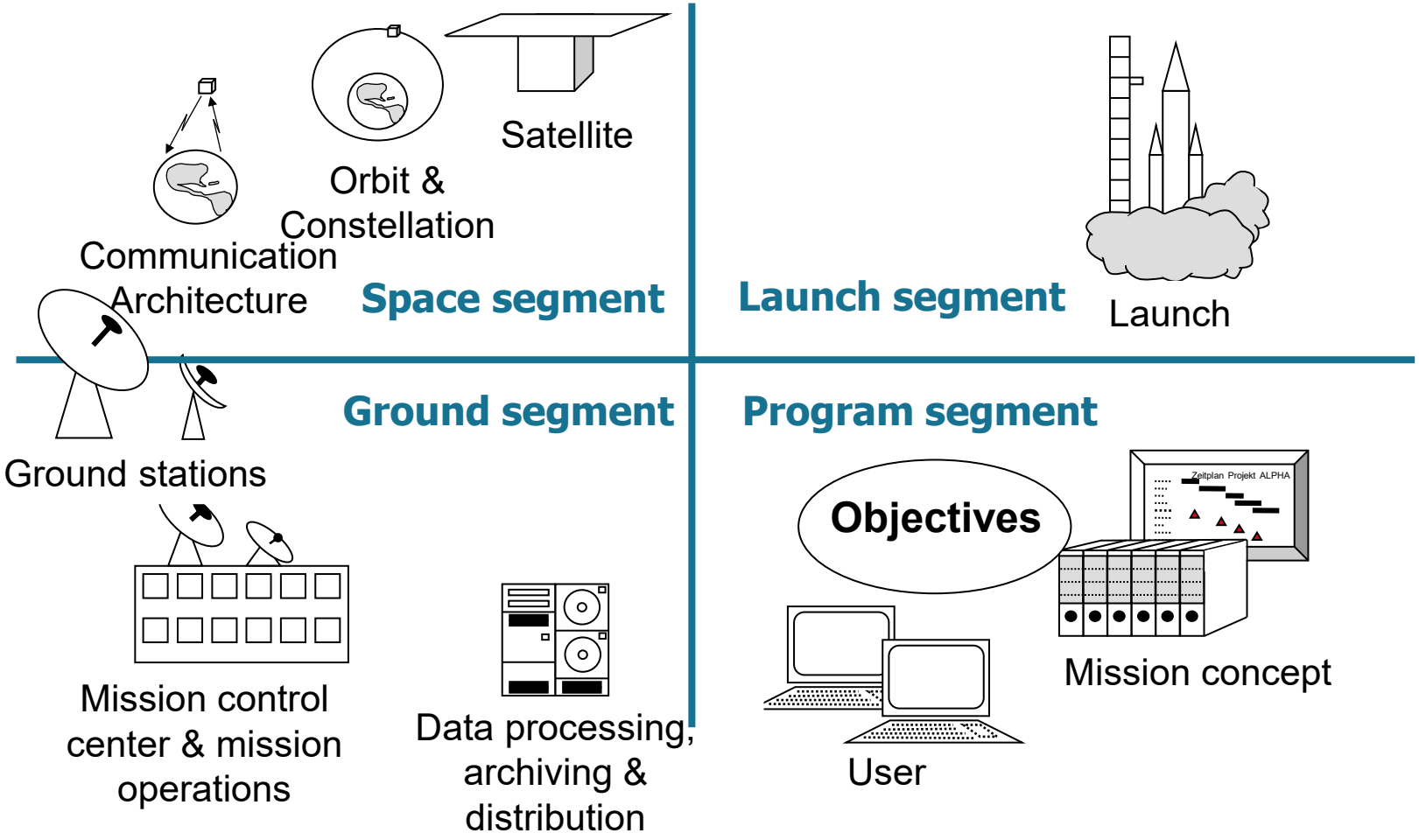
Tasks

- Define and iterate requirements
- Identify, analyse and set critical parameters for the mission

  ➔ Usually done by and/or with mission scientists
- Provide detailed information on all external (esp. environmental) aspects of a mission
- Provide reports, calculations and simulations before and after the mission start
  - Trajectory propagations and optimisation
  - Maneuver plan, navigation strategies
  - Visibility, contact and eclipse analysis

Mission Design Process

Segments of a satellite mission



Mission statement

Should answer three questions:

1. Why do you want to do it?
2. What do you want to do?
3. Who should profit from it?

➔ Basis for all further considerations, planning etc.

Mission objectives

- Based on statement
- Qualitative goals of the mission

  ➔ formulated in cooperation with stakeholders, developers, operators

  ➔ common grounds for further mission definition
- Can be devided into primary and secondary
- In general not changed afterwards, may be sligthly adapted

User needs

- There are three different user groups
  - Scientific user
  - Commercial user
  - Governmental user

Mission requirements and constraints

Requirements

- Quantitative and qualitative
- Functional vs.operational requirements

Constraints

- not requirements
- Not derived from mission objectives

Mission concepts and alternative architectures

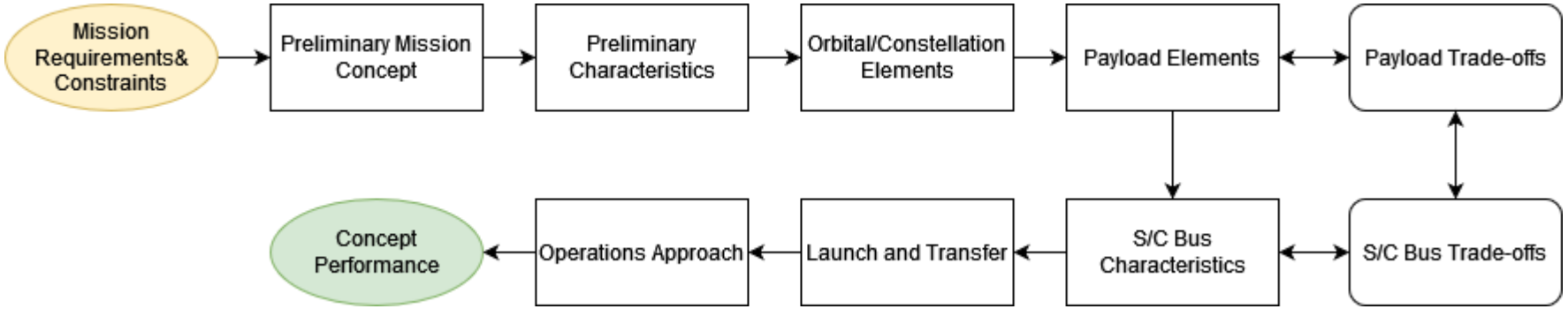
- Development of different mission concepts
  - ➔ all concepts have to fulfill mission objects and satisfy constraints
  - ➔ might offer different trade-offs
- Process might show that given the constraints only one architecure fully fulfills mission objectives
- Alternative mission concepts are developed/considered in phases 0 and A only

System drivers

- Main mission parameters determining the performance, cost, risk, schedule
- Can be controlled by user and/or designer
- Drivers are limited by some parameters and limite other parameters themselves
- Have to be defined for each alternative concept

Selection and Characterization of mission architecture

- Criteria for selection have to be identified and coordinated with all stakeholders
- Based on objectives and drivers
- Chosen solution must be described in detail for all parties



Critical Requirements

- Requirements dominating overall design

  ➔ strongly affecting costs and performance

  ➔ do not confuse with drivers

- Some questions to answer:
  - Which requirements substantially influence the system design and which are most difficult to fulfil?
  - What kind of accuracy is required for the tasks?
  - What are the expected costs?
  - What kind of operational requirements determine the system design and why?

Analysis and evaluation

- Different aspects have to be analyzed
  - Feasibility
  - Sizing
  - Trade-offs
  - Performance
  - Utility
  - ➔ Definition of criteria is crucial
  - ➔ Documentation is substantial part of it

Final Architecture

- Includes an argumentation for the chosen concept based on
  - Met objectives
  - Feasibility and utility
  - Risk accessment
  - Budget and schedule
  - Political constraints/considerations

Image index

[1] All rights reserved. http://www.raumfahrttechnik.tu-berlin.de

[2] https://www.esa.int/ESA_Multimedia/Images/2021/02/ ESOC_Mission_Control_Room

[3] http://www.space-systems.eu/index.php/de/forschung/projekte/74-category-en-gb/science/projects/201-innocube